MD Shihabul Islam Shovo

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Professional Summary

Computer Science and Engineering graduate with experience in AI research and full-stack software development. Currently working as a Research Assistant at ELITE Research Lab and a Web & Automation Engineer at Cleaning Connected, alongside research activities at NIMISHES Lab. Research experience includes developing and evaluating deep learning models, conducting experiments, and contributing to publications in computer vision, medical imaging, and related areas. Software development experience includes building web applications, REST APIs, and automation systems using Python, Django, React, and ASP.NET.

Research Interests

Medical Image Analysis · Lightweight & Efficient Deep Learning · Semi-Supervised Learning · State-Space Models (Mamba) · Biosignal & Wearable Sensing · Explainable AI · Parameter-Efficient Adaptation · Trustworthy Vision-Language Models

Education

B.Sc. in Computer Science and Engineering — *2020 – 2024*
Ahsanullah University of Science and Technology (AUST) — *Dhaka, Bangladesh*

Higher Secondary Certificate (HSC) — *2016 – 2018*
Cantonment Public School & College, Saidpur — *Rangpur, Bangladesh*

Secondary School Certificate (SSC) — *2014 – 2015*
Cantonment Public School & College, Saidpur — *Rangpur, Bangladesh*

Technical Skills

AI/ML & Research	PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, Mamba / state-space models, Grad-CAM & explainable AI, multi-seed statistical validation (Wilcoxon, ANOVA), structural reparameterization
Foundation Models	LoRA / parameter-efficient fine-tuning, vision transformers (DINOv2, CLIP, SAM), vision-language models (Qwen2.5-VL, LLaVA-Med), BiomedCLIP retrieval, tool-using (ReAct) agents, linear CKA representation analysis
Languages	Python, JavaScript, TypeScript, C#, Java, C/C++
Frontend	React, Tailwind CSS, Bootstrap, HTML, CSS
Backend	Django, Django REST Framework, ASP.NET MVC, Flask, PHP
Databases	SQLite, MySQL, PostgreSQL, Microsoft SQL Server
Tools & Auth	Git, GitHub, Postman, Selenium WebDriver, JWT Authentication, Swagger/OpenAPI
Concepts	REST APIs, MVC Architecture, RBAC, Data Structures & Algorithms, SDLC, Agile Methodology

Professional Experience

Web & Automation Engineer, *Cleaning Connected* — *Jan 2025 – Present*
California, USA
(Remote) · Part-time

- Designed and developed digital solutions supporting business operations, including website development, lead management automation, and workflow optimisation.
- Maintained web platforms, integrated business tools, and resolved technical issues to improve operational efficiency.

Research Experience

ELITE Research Lab LLC — *Jun 2026 – Present*
Research Assistant — *Queens, NY, USA*
(Remote) · Part-time

Project: LCM-UNet – Lightweight Skin Lesion Segmentation using a Mamba-based U-Net

- Designed and led development of an ultra-lightweight (∼34K-parameter, 0.06 GFLOPs) Mamba-based state-space U-Net for melanoma-related lesion boundary detection, targeting deployment on low-cost point-of-care hardware.
- Proposed the Residual Dilated Local Adapter (RDLA), a zero-initialized dilated convolution module that adds optimisation capacity during training and is exactly fused back into the base kernel via structural reparameterization before inference, at zero added deployment cost.
- Validated the approach with multi-seed statistical testing (image-level Wilcoxon, $p < 10^{-4}$) on ISIC 2017, ISIC 2018, and external PH2 datasets, demonstrating consistent boundary-quality improvements over the baseline.

NIMISHES Lab, *Lead Researcher* *2026 – Present*

- Independent research lab building explainable, deployment-efficient AI for clinical use; contributed to 8 manuscripts across medical imaging (dermatology, ophthalmology, radiology), wearable biosignal analysis (EEG, IMU), and underwater computer vision, including first-author AquaProbe with the University of South Florida and APED-CXR with the American University of Sharjah.
- Extended the lab's work into parameter-efficient adaptation of frozen foundation models (LoRA rank allocation for vision transformers) and the reliability of agentic vision-language systems for medical imaging.
- Contributed to model design, cross-validation and statistical significance testing, and Grad-CAM-based explainability analysis across projects, consistently achieving state-of-the-art parameter efficiency versus standard backbones (e.g. ResNet50, DenseNet201).

Software Development Projects

Hospital Management System, *Full Stack (DRF, React, JWT, SQLite/PostgreSQL)* GitHub

- Designed and built a role-based hospital management platform (admin, doctor, patient, receptionist) with a Django REST Framework API and a React (Vite) SPA, modeling 9 relational entities including patients, doctors, appointments, prescriptions, and billing.
- Implemented JWT authentication with automatic silent token refresh (with request deduplication) and dual-layer role-based authorization that combines queryset-level scoping with object-level permission checks, ensuring users can only access their own records.
- Built a consistent API error-handling layer (custom DRF exception handler) with auto-generated OpenAPI/Swagger docs via drf-spectacular, reusable React data-fetching hooks, protected/role-gated routing, and a centralized axios client with request/response interceptors.

Multi-Tenant SaaS Project Management Platform, *Django REST Framework, PostgreSQL, Celery, Redis, Docker, JWT, React* GitHub

- Architected and developed a scalable multi-tenant project management API using Django REST Framework with organization-level data isolation, JWT authentication, and role-based permissions for administrators, managers, and team members.
- Implemented asynchronous background task processing with Celery and Redis for notifications, activity tracking, and scheduled operations, decoupling long-running tasks from the request cycle.
- Optimized database performance using Django ORM query optimization, strategic indexing, cursor-based pagination, and efficient relational modeling on PostgreSQL with support for millions of records.

EasyHome: House Rentals Service, *ASP.NET MVC, C#, Entity Framework, MSSQL, Bootstrap* GitHub

- Built a complete house rental management system for landlords and tenants with role-based access, apartment dashboards, and rent requests.
- Implemented reviews, ratings, and an in-app chat system alongside user registration and profile management.

MRI Tumor Detection System, *Python, Flask, TensorFlow, Keras, VGG16* GitHub

- Developed a web-based AI system to classify MRI brain scans into four tumor categories.
- Implemented confidence scoring and automated PDF medical report generation, and optimized a CNN-based model for medical image classification.

NOTES App, *Django REST Framework, React, SQLite, JWT* GitHub

- Implemented user-specific CRUD operations with protected routes, integrating frontend and backend via REST APIs and CORS configuration.

Publications

Accepted (4) IEEE conferences and IEEE/CVF workshops

[C1] AquaProbe: Efficient Adaptation of Frozen Vision Models for Underwater Images
M. S. I. Shovo, N. K. Paul, T. Dong, and X. Lin *(First author)*
5th Maritime Computer Vision Workshop (MaCVi), IEEE/CVF WACV 2027 arXiv
70.1 mIoU on SUIM with a 0.59M-parameter adapter budget; +1.9 mIoU over uniform LoRA rank allocation.

[C2] WaveFoG: Wavelet-Gated Transformer for Parkinson's Freezing of Gait Detection Using Wearable Accelerometer Signals
M. S. I. Shovo, N. K. Paul, A. Rahman, and I. J. Esha *(First author)*
5th IEEE BECITHCON 2026, Dhaka, Bangladesh arXiv · Code
F1 0.875 ± 0.017 on the Kaggle TLVMC FoG dataset under subject-independent grouped 10-fold CV.

[C3] SleepEffFormer: Efficient CNN-Transformer with Transition-Aware Smoothing for Single-Channel EEG Sleep Stage Classification
N. K. Paul, **M. S. I. Shovo**, I. J. Esha, and A. Rahman
5th IEEE BECITHCON 2026, Dhaka, Bangladesh arXiv · Code
83.9% accuracy and 78.9% macro-F1 on Sleep-EDF with ∼367K parameters.

[C4] DR-LiteNet: A Lightweight Explainable Hybrid CNN Framework for Imbalanced Diabetic Retinopathy Grading via Adaptive SMOTE Fusion
NIMISHES Lab
1st IEEE iCONEECT 2026 arXiv
88.77% accuracy and QWK 0.884 on APTOS 2019 with 4.69M parameters.

Under Review (5) 1 IEEE JBHI, 3 IEEE ISBI 2027, 1 IEEE/CVF WACV 2027 FABRIC

[U1] LCM-UNet: A Reparameterizable Local-Compensated Mamba U-Net for Skin Lesion Segmentation
M. S. I. Shovo, N. K. Paul, and K. Morol *(First author)*
IEEE Journal of Biomedical and Health Informatics (JBHI) arXiv
34K parameters and 0.06 GFLOPs; validated on ISIC 2017/2018 and external PH2 (Wilcoxon $p < 10^{-4}$).

[U2] APED-CXR: Adaptive Privileged Evidence Distillation for Chest X-Ray Grounding with Image-Only Inference
M. S. I. Shovo, N. K. Paul, and J. Fayyad *(First author)*
IEEE International Symposium on Biomedical Imaging (ISBI) 2027 arXiv
11.1M-parameter image-only student; PadChest-GR micro-gF1 improved from 0.302 to 0.338.

[U3] Band2Net: Frequency-to-Network Transport Distillation for fMRI-Privileged, EEG-Only Session-Condition Decoding
N. K. Paul, **M. S. I. Shovo**, and D. Mahapatra
IEEE International Symposium on Biomedical Imaging (ISBI) 2027 arXiv
EEG-only ROC-AUC 0.801 on PsiConnect; +5.0 AUC points over an EEG-only student.

[U4] LAPIS: Clinically Constrained Patient-Identity Residual Learning for Chest X-Ray Anonymization
N. K. Paul, **M. S. I. Shovo**, and D. Sidibé
IEEE International Symposium on Biomedical Imaging (ISBI) 2027 arXiv

[U5] Coupled or Independent? Fault Coupling in Agentic Medical Vision Systems
N. K. Paul, **M. S. I. Shovo**, M. M. Rahman, and J. Fayyad
FABRIC Workshop, IEEE/CVF WACV 2027 arXiv
Tool order changes hidden joint failures: 55% versus 35% wrong agreement.

Thesis & Independent Research

Study on Benchmark Datasets for Human Disease Genes, *Academic Research*

- Analyzed gene–disease associations using benchmark biomedical datasets including DisGeNet, OMIM, CTD, and GWAS.
- Evaluated machine learning models including Decision Trees, Random Forests, SVM, and KNN for classification performance, and explored genomic data extraction tools including BioBERT and PubTator.

Certifications and Professional Development

Artificial Intelligence Technologies 2026 Summer School, *Ivan Franko National University of Lviv* *Jun – Jul 2026*
120 hours / 4 ECTS credits. Focus: Machine learning, Python for data analysis, deep learning architectures, computer vision, cloud computing, and AI applications. Certificate

Servercamp, *Astha IT* *May 2026 – Ongoing*
Focus: Backend development (4-month program) with C#, OOP, ASP.NET, and SQL.

Full Stack Web Development with Python, Django & React, *Ostad* *Oct 2025 – Completed*
Focus: JavaScript, Python, HTML, CSS, Django, and React. Certificate

Competitive Programming & Extracurriculars

</> Codeforces · </> LeetCode

- Participated in Intra AUST Programming Contest, Fall 2021.
- Campus Ambassador, GUVI Geek Networks (IITM Research Park), from 2021 to 2023; organized a seminar on the importance of Data Science.